Business Statistics Solutions

Question Two

Differentiate between mutually exclusive events and mutually non-exclusive events. Give examples.

Mutually Exclusive Events:

- These are events that cannot occur at the same time.
- If one event happens, the other must not happen.
- Mathematically: P(A and B) = 0
- Example: Tossing a coin - getting a head or a tail (not both). Drawing a card - getting a King or a Queen.

Mutually Non-Exclusive Events:

- These are events that can occur simultaneously.
- One event occurring does not prevent the other from occurring.
- Mathematically: P(A and B) != 0
- Example: Drawing a card - getting a red card and a King. A student passing both Math and English.

Question Three

Explain five problems faced while constructing an index number.

1. Selection of Base Year:
   - Choosing a base year that is free from economic fluctuations is difficult.

2. Selection of Items:
   - Deciding which goods or services to include can be subjective and may not reflect actual consumption.

3. Choice of Weighting:

- Assigning appropriate weights to different items is challenging and affects the accuracy of the index.

4. Price and Quality Changes:

- Index numbers may fail to reflect changes in quality or brand, only tracking price changes.

5. Data Collection Problems:

- Accurate and timely data is often difficult to collect, especially in developing areas.

Prepared by ChatGPT